

Supplementary Figures: Strain–Doping Coupling in Graphullerene

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Companion to the main manuscript on coupled (ϵ, δ) response in qHP C₆₀. Figures follow the observables vocabulary of Capobianco *et al.* [1] (localization, λ ,

J , FCWD). **DATA:** Fig. S5 Marcus λ panels refresh from `exp9.polaron_verification.json` after vertical SP converge; until then literature λ schematics only.

[1] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa,

Nano Letters **24**, 7891 (2024), key reference from refer-info.md - polaron transport mechanisms.

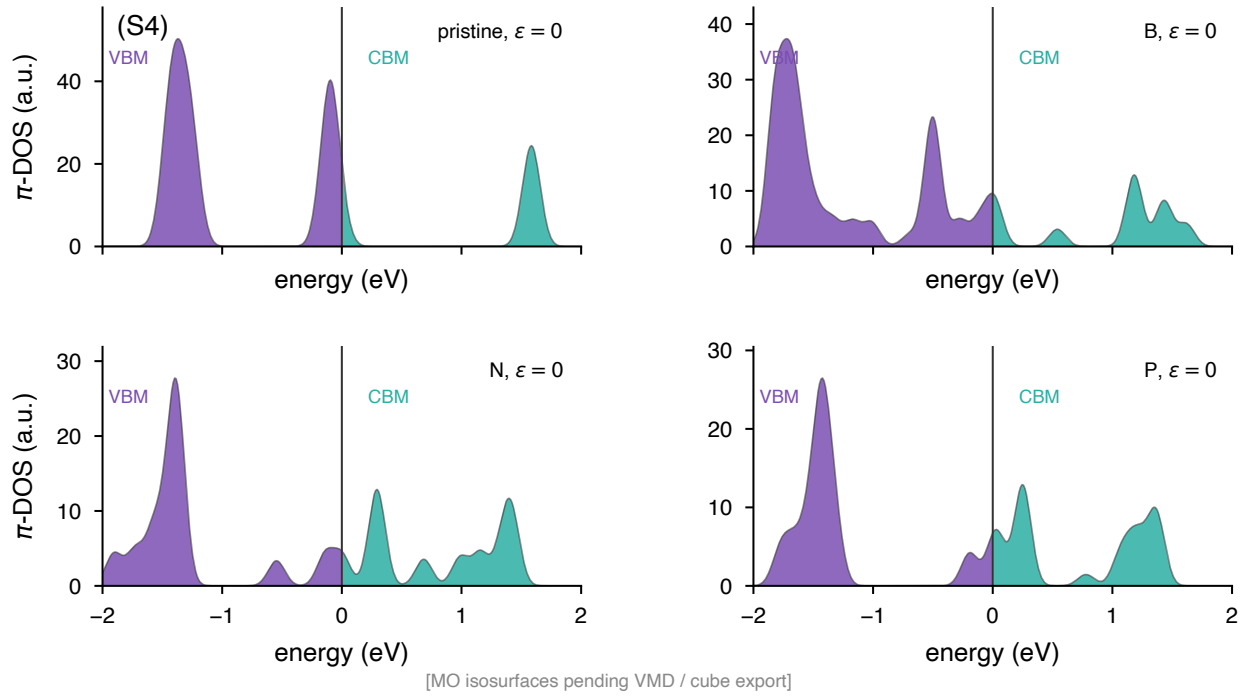


FIG. 1. π -projected DOS at $\varepsilon=0$ for pristine and B/N/P tetramers (Exp. 7 .pdos, PBE+D3). MO isosurfaces [pending: VMD + cube export]; band-edge labels follow E_F .

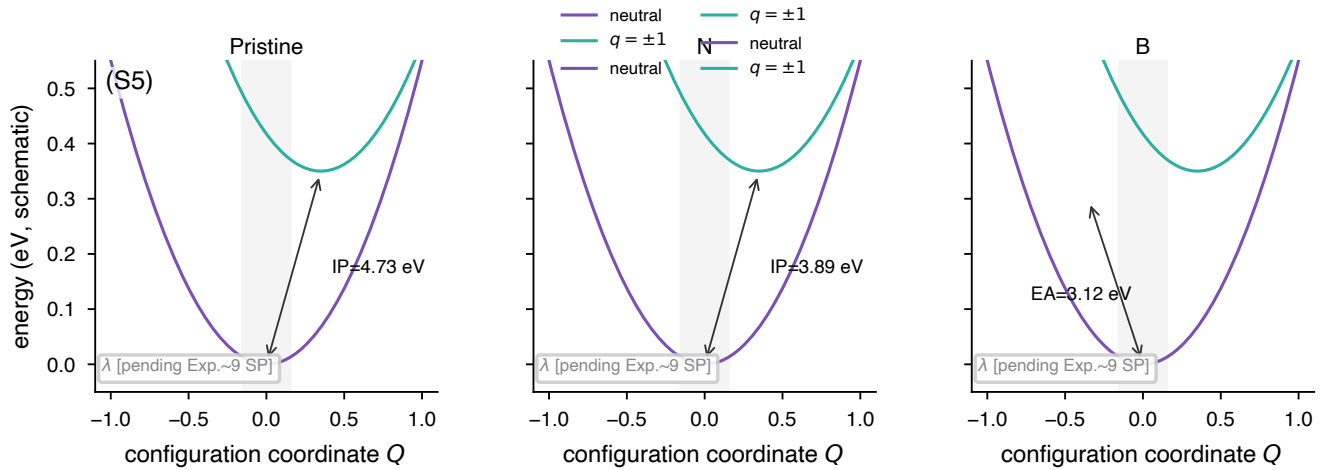


FIG. 2. Marcus configuration-coordinate schematic (Capobianco-style). Verified adiabatic IP/EA from converged Exp. 9 GEO-OPT pairs label the vertical arrows (pristine IP, N IP, B EA); λ panels remain [pending: vertical SP at neutral geometry]—see `derived.lambda.eV` in audit JSON.

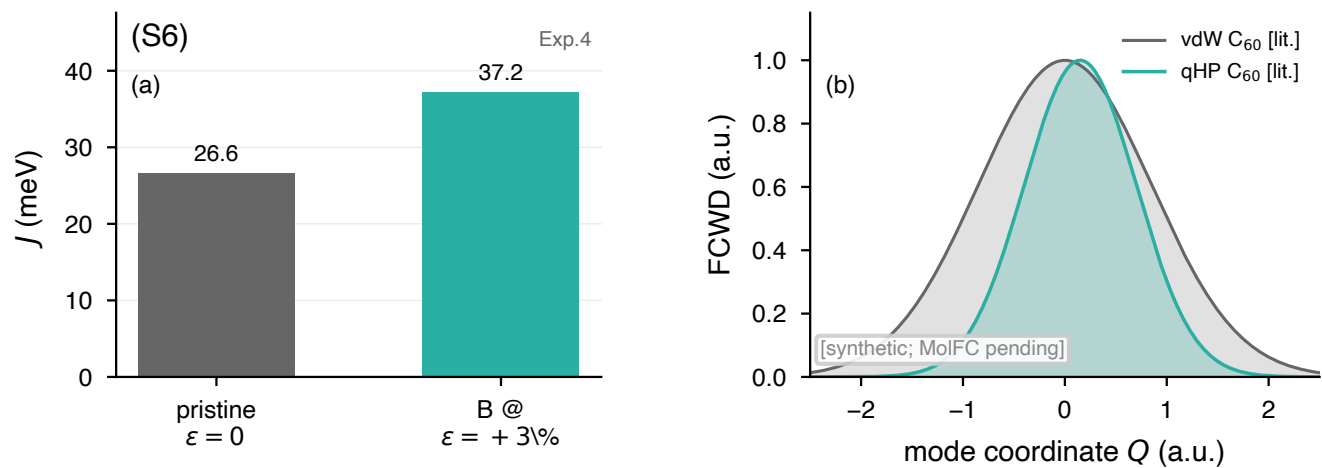


FIG. 3. Transport-adjacent observables: (a) nearest-neighbor J from Exp. 4 audit (PBE+D3, two verified tetramer points); (b) illustrative Franck-Condon weighted density (FCWD) envelope following Ref. 1 [**synthetic; MolFC replacement pending**].